

MIREL AGY FERNANDES

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PROFESSIONAL SUMMARY

Data Engineer with 5+ years of experience designing scalable data platforms and distributed ETL/ELT pipelines using Apache Spark, Snowflake, Azure, and dbt. Proven track record processing large-scale data in cross-functional, client-facing environments across healthcare, finance, and tech. Passionate about enabling ML-ready data infrastructure and delivering measurable business impact.

TECHNICAL SKILLS

Languages: Python, SQL, Scala, PySpark, Snowpark | **Big Data:** Apache Spark, Hadoop, Databricks, MapReduce **Cloud & Warehousing:** Azure (ADLS Gen2, Synapse, ADF, Databricks), Snowflake, BigQuery, Delta Lake **ETL & Orchestration:** dbt, Azure Data Factory, Apache Airflow, Kestra, SSIS **Data Quality:** dbt tests, Great Expectations, automated validation, PII/GDPR compliance **DevOps:** Docker, Kubernetes, Terraform, Git, CI/CD, Azure DevOps | **BI:** Power BI, Qlik, SSRS **Analytics & ML Libraries:** pandas, NumPy, spaCy, NLTK, TensorFlow, BERT, SHAP, scikit-learn

PROFESSIONAL EXPERIENCE

Data Engineer | Imperial College Healthcare NHS Trust / Imperial College London

March 2024 - Present | London, United Kingdom

- Designed and deployed scalable ETL/ELT pipelines extracting EHR data from 4 source systems (Cerner, Arya Radiology, Somerset Cancer, Nautilus BLOB) into Snowflake via Snowpark, supporting 20+ research projects and enabling ML-ready data infrastructure.
- Engineered a clinical notes anonymisation pipeline using Apache Spark and a BERT-based transformer (MMIC-4 standard), achieving 99.9% de-identification accuracy on millions of records.
- Built an automated data validation framework (Snowpark) validating 180M records (20 columns) in 1 minute with 100% PII leakage prevention, ensuring ML-ready data quality.
- Authored dbt transformation logic, test cases, and macros supporting both full and incremental loads, applying data quality checks and automated regression testing as part of the CI/CD pipeline.
- Developed analytical models using pandas, spaCy, NLTK, and NumPy as the basis for Qlik and Matplotlib visualisations delivered to Data Scientists and research teams for data verification.
- Leveraged Azure ML with GPT-4.0 API to extract insights from Snowflake data for automated discharge summary generation.
- Enriched structured fields using Snowflake Cortex, increasing population from 60% to 90%; developed linkage models achieving 98% accuracy.
- Collaborated with clinical researchers, Data Scientists, data protection officers, and senior stakeholders to translate business and research requirements into

technical solutions and measurable outcomes.

Senior System Developer / Azure Data Engineer | Infosys Limited

November 2019 - April 2022 | Bangalore, India (Clients: MetLife, Citi, Apple)

- Delivered data engineering solutions in a fast-paced consulting model, managing technical delivery and stakeholder communication across enterprise clients (MetLife, Citi, Apple) from requirements to production.
- Led legacy-to-cloud migration using Azure Data Factory, Databricks, and PySpark/Scala, delivering 40% reduction in processing time.
- Built Bronze layer ingestion pipelines in Azure Databricks into ADLS Gen2 with schema enforcement and incremental loading; contributed to reusable cross-client asset libraries.
- Optimised 15 SQL stored procedures and 215+ tables, reducing process time by a further 20%.
- Led an Agile team of 4 engineers; rated 4.5/5 in appraisals and earned High Performer Award.

PROJECTS

- **End-to-End Data Engineering Portfolio (2025-2026):** Reusable, production-grade pipeline templates — Docker, Kestra/Airflow, BigQuery/GCP, Spark, dbt, Terraform — designed as shareable internal assets. github.com/Mirel27/data-engineering-zoomcamp
- **NLP Depression Diagnosis Pipeline (MSc, 2023):** ML feature engineering and fine-tuning of BERT/MentalRoBERTa on Reddit & YouTube data; built end-to-end data pipeline for ML classification; F1 = 0.967; SHAP interpretability analysis. Python, HuggingFace, PyTorch.

EDUCATION

MSc Data Science — University of Sheffield | 2022 - 2023 **B.E. Electronic & Telecommunication Engineering** — Goa University | 2015 - 2019

CERTIFICATIONS

Snowflake Data Engineer (2025) | dbt Fundamentals (2025) | Azure Data Engineer Associate DP-203 (2023) | Azure AI Fundamentals AI-900 (2023)

ACHIEVEMENTS & LANGUAGES

- 2nd place — GitHub Copilot Hack Tour (data profiling PoC)
- English: Fluent / Professional